Improving Driver Efficiency and Earnings on Uber through Demand Visibility

1. Problem Statement

Ride-hailing platforms such as Uber operate in a two-sided marketplace where maintaining an optimal balance between driver supply and rider demand is critical. A key challenge in this ecosystem is **driver retention**, which is heavily influenced by the consistency and predictability of driver earnings.

Preliminary analysis indicates that a significant proportion of drivers experience **inconsistent earnings and high idle time**, despite remaining online for extended durations. This inefficiency arises due to **demand-supply mismatches, suboptimal driver positioning, and lack of real-time visibility into high-demand zones**.

As a result, drivers are unable to maximize their ride frequency, leading to **reduced earnings per hour and overall dissatisfaction**. Over time, this contributes to increased **driver churn**, which negatively impacts platform performance by reducing driver availability, increasing rider wait times, and ultimately affecting revenue generation.

This project aims to analyze driver performance data to identify the underlying causes of inefficiency and propose product-level interventions that improve **driver utilization, earnings stability, and platform reliability**.

2. Data & Methodology

To analyze the problem, a structured dataset was created simulating driver activity on a ride-hailing platform such as Uber.

The dataset consists of **50 drivers over a 10-day period**, resulting in approximately **500 data points**. Each record represents a driver's daily activity and includes the following variables:

- **driver_id** – Unique identifier for each driver
- **date** – Day of activity
- **hours_online** – Total time the driver was active on the platform
- **rides_completed** – Number of rides completed per day
- **earnings** – Total daily earnings (in INR)
- **city** – Location of operation

The dataset was generated to reflect realistic variations in driver behavior, including differences in working hours, ride frequency, and earnings patterns.

Analytical Approach

The analysis was conducted using SQL-based queries to evaluate driver performance across key dimensions:

- **Earnings efficiency** (earnings per hour)
- **Driver productivity** (rides per hour)
- **Performance distribution across drivers**

This approach enabled identification of performance gaps and inefficiencies at an individual driver level, forming the basis for further insights and product recommendations.

3. Data & SQL Analysis

To evaluate driver performance and identify inefficiencies, a structured dataset was used to simulate real-world activity on a ride-hailing platform such as Uber. The dataset includes key operational metrics such as hours online, rides completed, and total earnings for each driver across multiple days.

The dataset can be accessed here:

[]

driver_data (1).csv

SQL-based analysis was conducted to examine patterns in driver productivity and earning efficiency. Key metrics such as **earnings per hour** and **rides per hour** were derived to assess performance at both aggregate and individual levels.

This approach enables identification of performance disparities across drivers and provides a quantitative foundation for diagnosing the underlying causes of inefficiencies within the system.

4. SQL Setup & Data Preview

To enable structured analysis, the dataset was loaded into an in-memory SQL database using Python. This allowed efficient querying and aggregation of driver-level metrics.

Data Loading Process

The dataset was imported using `pandas` and stored in a temporary SQL table for analysis:

```
importpandasaspd
importsqlite3

# Load dataset
df=pd.read_csv("driver_data.csv")

# Create database in memory
conn=sqlite3.connect(':memory:')
df.to_sql('drivers',conn,index=False,if_exists='replace')

# Preview
df.head()
```

Sample Data Preview

A snapshot of the dataset is shown below:

driver_id	date	hours_online	rides_completed	earnings	city
D1000	2024-01-01	11.6	18	2194	Delhi
D1000	2024-01-02	5.2	5	420	Delhi
D1000	2024-01-03	10.9	15	1943	Bangalore
D1000	2024-01-04	4.2	7	967	Mumbai
D1000	2024-01-05	5.7	6	557	Mumbai

5. SQL Analysis: Earnings Efficiency

To evaluate driver-level earning performance, earnings per hour was calculated using aggregated earnings and total hours online.

Query

```
SELECT
    driver_id,
    SUM(earnings)*1.0/ SUM(hours_online)AS earnings_per_hour
FROM drivers
GROUPBY driver_id
ORDERBY earnings_per_hourASC
LIMIT10;
```

Output (Lowest Performing Drivers)

driver_id	earnings_per_hour
D1019	106.31
D1020	110.55

driver_id	earnings_per_hour
D1014	112.41
D1029	112.45
D1042	115.13
D1030	119.14
D1046	120.06
D1022	123.33
D1005	124.15
D1001	126.77

Key Findings

The analysis reveals a significant variation in earnings efficiency across drivers. The lowest-performing drivers earn between **₹106 and ₹126 per hour**, which is substantially lower than the overall average (approximately ₹165 per hour).

This represents a gap of nearly **25–35% in earnings efficiency**, indicating that a segment of drivers is unable to generate sustainable income despite being active on the platform.

Insight

The disparity in earnings suggests underlying inefficiencies in how driver time is utilized. Since earnings are directly dependent on ride completion, lower earnings per hour point toward reduced ride frequency or increased idle time.

This finding establishes a strong link between **earnings inconsistency and potential driver dissatisfaction**, making it a critical factor contributing to driver churn.

6. SQL Analysis: Aggregate Performance Overview

To establish a baseline understanding of overall driver performance, aggregate metrics were calculated across all drivers.

Query

```
SELECT
    AVG(earnings)AS avg_earnings,
    AVG(hours_online)AS avg_hours,
    AVG(rides_completed)AS avg_rides
FROM drivers;
```

Output

avg_earnings	avg_hours	avg_rides
1328.39	8.07	11.85

Key Findings

On average, drivers earn approximately **₹1328 per day** while being online for around **8.07 hours**, completing **11.85 rides per day**.

This translates to:

- **~₹165 earnings per hour**
- **~1.47 rides per hour**

Insight

While the average performance appears moderate, these aggregate values mask significant variation across individual drivers. The relatively low rides per hour (~1.47) suggests that drivers spend a considerable portion of their time idle between trips.

This indicates that **driver time is not being utilized efficiently**, which directly impacts earnings potential. Given that drivers remain active for extended hours, the gap between effort (time spent) and output (rides completed) highlights a structural inefficiency in the system.

Implication

This baseline establishes a reference point against which low-performing drivers can be compared. Subsequent analysis reveals that a segment of drivers performs significantly below these averages, reinforcing the presence of inefficiencies that contribute to inconsistent earnings and potential churn.

7. Combined Analysis: Driver Efficiency & Earnings Disparity

The analysis of aggregate and driver-level metrics reveals a clear pattern of inefficiency in driver performance.

At an overall level, drivers earn approximately **₹165 per hour** and complete around **1.47 rides per hour**. However, a closer examination of individual performance highlights a significant deviation from these averages.

The lowest-performing drivers earn between **₹106 and ₹126 per hour**, representing a **25–35% reduction** compared to the average. This disparity indicates that a segment of drivers is unable to achieve optimal earnings despite being active on the platform.

Further analysis of ride frequency shows that these drivers also exhibit lower operational efficiency, completing significantly fewer rides per hour. This directly impacts their earning potential, as income is strongly correlated with ride volume.

Key Insight

The findings establish a direct relationship between:

> **Lower ride frequency → Increased idle time → Reduced earnings per hour**

This indicates that the primary issue is not insufficient working hours, but rather **inefficient utilization of driver time**.

Root Cause Identification

The observed inefficiencies can be attributed to:

- Demand-supply mismatch across locations
- Lack of real-time visibility into high-demand zones
- Suboptimal driver positioning

As a result, drivers spend a significant portion of their active time waiting for ride requests, leading to inconsistent earnings and reduced overall satisfaction.

Conclusion

The analysis confirms that **earnings inconsistency is a symptom of deeper inefficiencies in ride allocation and demand visibility**. Addressing these inefficiencies presents a clear opportunity to improve driver productivity, stabilize earnings, and reduce churn on the platform.

8. SQL Analysis: Driver Efficiency (Rides per Hour)

To further investigate the drivers of earnings disparity, driver efficiency was measured using rides completed per hour.

Query

```
SELECT
    driver_id,
    SUM(hours_online)AS total_hours,
    SUM(rides_completed)AS total_rides,
    SUM(rides_completed)*1.0/ SUM(hours_online)AS rides_per_hour
FROM drivers
GROUPBY driver_id
ORDERBY rides_per_hourASC
LIMIT10;
```

Output (Lowest Performing Drivers)

driver_id	total_hours	total_rides	rides_per_hour
D1029	82.1	78	0.95
D1014	92.2	88	0.95
D1042	86.5	88	1.02
D1012	82.2	84	1.02
D1020	85.0	88	1.04
D1046	83.9	87	1.04
D1022	75.1	81	1.08
D1001	81.1	88	1.09
D1030	75.9	83	1.09
D1019	74.2	82	1.11

Key Findings

The lowest-performing drivers complete between **0.95 and 1.10 rides per hour**, compared to the overall average of approximately **1.47 rides per hour**.

This represents a reduction of nearly **25–35% in ride frequency**, indicating significantly lower operational efficiency.

Insight

The reduced rides per hour strongly suggests that these drivers experience **higher idle time between trips**. Despite being online for substantial durations (75–92 hours over the observed period), their ride completion remains comparatively low.

This inefficiency directly explains the previously observed **lower earnings per hour**, confirming that earnings disparity is primarily driven by differences in ride frequency rather than time spent online.

Conclusion

The analysis establishes a clear causal relationship:

> **Lower rides per hour → Increased idle time → Reduced earnings per hour**

This reinforces the conclusion that the core issue lies in **inefficient utilization of driver time**, rather than lack of effort or insufficient working hours.